Debre Berhan University

College of Computing

Department of Computer Science

Internship Report

Project Title: Exhibition Management System

Hosting Company: eTech S.C

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September, 2025

Acknowledgements

We would like to extend our heartfelt gratitude to all those who contributed to the successful completion of this web development project during our internship at **eTech Sc**. Their collaborative efforts and invaluable support helps us a lot. We extend our sincere appreciation to everyone who guided us in translating academic theory into a functioning, real world application.

First, We express our deepest gratitude to Almighty God and His Mother Saint Mary. The health, perseverance, and focus required to successfully navigate this intensive internship and complete the associated documentation were true blessings, and We are thankful for the grace provided throughout this journey.

We are so grateful to **Debre Berhan University**, specifically the College of Computing Science and the Department of **Computer Science**. This structured internship served as the essential bridge between our theoretical knowledge and professional practice, providing the foundation for everything we achieved at eTech Sc.

A huge thank you goes to **eTech Sc**. for generously hosting our internship. They entrusted us with the development of the Exhibition or Conference Management System a task that required us to engage fully with their professional ecosystem. Their commitment to high standards coupled with the opportunity to use cutting-edge tools like Vert.x and Angular, was pivotal in refining our technical and problem solving skills.

Finally, We reserve special thanks for our direct mentors: Our University Mentor, Hiwot Kadi, and our Company Supervisor, Endeshaw Aweqe. Their dedicated guidance and belief in our ability were crucial in keeping the project on track and ensuring we gained maximum practical experience.

We owe a special note of gratitude to the professional developers who generously shared their time and expertise, particularly Mr. Robel and Mr. Yohannes. Their willingness to discuss advanced architectural concepts and offer constructive insights significantly enhanced the design and implementation of our project.

Finally, We must acknowledge our fellow interns from both Debre Berhan University and other institutions at eTech. The collaborative atmosphere we fostered, characterized by mutual support and sharing project ideas, was instrumental to our collective success and made the entire experience rewarding.

To our family and friends, thank you for your unwavering encouragement.

Declaration

We, Tsigereda and Netsanet, hereby formally declare the following regarding this project report, titled "Project on Web Development: Exhibition Management System": This document is the result of our own professional application and research during the internship period.We confirm that this report has not been submitted in any form for another degree, diploma or qualification at any other university or educational institution.All external sources of information, materials and support have been meticulously cited and acknowledged throughout the text.We take full and complete responsibility for the accuracy of the data, findings and conclusions presented within this report.
This report is submitted in partial fulfillment of the requirements for the Bachelor degree in Computer Science at Debre Berhan University.

Execuitive Summery

This internship project presents a comprehensive Exhibition Management System built using modern full-stack technologies to streamline the entire conference lifecycle from planning to execution. The system features a responsive Angular 20.1.0 frontend with TypeScript and Angular Material components, supported by a high-performance Java backend using the Vert.x framework (4.5.7) and PostgreSQL database. The platform serves multiple user roles including administrators, exhibitors, and attendees, with JWT-based authentication ensuring secure access control.

The system provides core event management capabilities, including exhibitor registration, booth-related workflows, payment processing through an integrated Stripe gateway, attendee and speaker data management, sponsor/partner tracking, and administrative monitoring. It also includes floor management and product catalog features to support exhibition logistics and exhibitor operations.

Email integration capabilities are implemented using Vert.x Mail Client (4.5.7) for automated notifications, welcome emails for new registrations, password reset functionality, and system communications, ensuring seamless user engagement and support throughout the conference management process. Built with scalability and security in mind, the application features normalized database design with proper relationships and constraints, reactive programming patterns for optimal performance, file upload capabilities for branding and documentation, and comprehensive validation throughout.

This project demonstrates proficiency in modern web development practices, API design, payment integration, email automation, database architecture, and user experience design, delivering a complete digital solution that automates conference operations, ensures secure financial transactions, provides automated communication workflows and offers intuitive interfaces for all stakeholders while maintaining data integrity and supporting multi-tenant operations.

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Acronyms

JWT - JSON Web Token

API - Application Programming Interface

REST - Representational State Transfer

CRUD - Create, Read, Update, Delete

HTTP - HyperText Transfer Protocol

JSON - JavaScript Object Notation

SQL - Structured Query Language

PK / FK - Primary Key / Foreign Key

RBAC - Role-Based Access Control

SPA - Single Page Application

CLI - Command Line Interface

CORS - Cross Origin Resource Sharing

MIME - Multipurpose Internet Mail Extensions

SDK - Software Development Kit

HS256 - HMAC with SHA-256

RDBMS - Relational Database Management System

DTO - Data Transfer Object

URL / URI - Uniform Resource Locator / Identifier

DDL / DML - Data Definition Language / Data Manipulation Language

UX / UI - User Experience / User Interface

Part Two: Introduction and Background of eTech SC

INTRODUCTION

eTech SC Technology and Business S.C is a leading Ethiopian technology solutions provider dedicated to empowering the nation's digital future. Established as a Share Company, eTech SC has positioned itself as a cornerstone of Ethiopia's technological landscape, specializing in the development of mission-critical software platforms and digital infrastructure. The company's work is integral to the daily operations of major financial institutions, the identity management of citizens, and the digital transformation of both public and private enterprises. eTech SC is driven by a clear strategic purpose and serves as the foundation for all its projects and internal culture.

❖ Mission of the Company

The mission of eTech S.C. is to **"create a borderless ICT expertise eco-system to do ethical and to end business anywhere."** This mission statement reflects the company's ambition to build a global network of technology expertise that operates without borders, all while adhering to the highest ethical standards to facilitate business operations seamlessly and integrity worldwide.

❖ Vision of the Company

eTech S.C. envisions itself as **"a pioneer in providing integrated Ethical Technology Solutions in Ethiopia and a noticeable player in the world by 2030."** This vision underscores a dual focus: achieving domestic leadership by being the foremost provider of integrated, ethical tech solutions within Ethiopia, while simultaneously establishing a significant and respected presence on the global stage within the next decade.

❖ Company Goals and Core Values

eTech S.C.'s strategic direction is built upon objectives designed to guarantee sustained market leadership, growth, and a measurable positive contribution to society. The primary goals are to:

- **Create Demonstrable Client Value**: Consistently produce outcomes that offer justifiable and measurable value for every client engagement.
- **Build a Respected Brand**: Position eTech as the standard for quality, reliability, and innovation within the ICT industry.
- **Maximize Shareholder Returns**: Ensure the company delivers sustainable financial returns and optimizes value for its investors.
- **Contribute to Society**: Actively fulfill corporate social responsibility by leveraging technology to achieve positive impacts on the community.

These overarching objectives are firmly supported by core values, including Quality, Customer Satisfaction, Loyalty, Trust, Integrity, Technological Excellence, and Empowering Partners.

2.1. Background

eTech SC is deeply intertwined with Ethiopia's recent digital transformation journey. The company was founded by a group of visionary professionals who identified a critical gap in the market for sophisticated, locally-developed technology solutions that understand the unique context of the Ethiopian business and regulatory environment. eTech S.C. stands out as a visionary leader in Ethiopian technology, focused on delivering comprehensive, end-to-end Information and Communication Technology (ICT) solutions. The company operates on the foundational principles of being Ethical, Ethiopian and End-To-End, driven not just by technological excellence, but by a deep commitment to humanity and meeting the nation's unique needs through local talent and innovation.

Demonstrating significant operational strength, eTech commands a capital base of 200 Million ETB and supports over 40 clients across diverse sectors with its growing team of more than 75 employees. A key differentiator is its governance: a distinguished board of executive management and shareholders features over 25 experts who bring international experience from more than 20 countries (including the USA and Europe) in fields like business, cybersecurity, and finance. This powerful convergence of local insight and global expertise uniquely positions eTech to spearhead digital transformation within Ethiopia and beyond.

2.2. Main Products, Services and Technologies

Their key offering is a comprehensive **Enterprise Resource Planning (ERP)** system, designed to integrate and manage the core business processes of large organizations, from finance and human resources to supply chain and operations. This product highlights the company's capacity to deliver complex, enterprise-wide software that enhances operational efficiency and data driven decision making for its clients. Furthermore, eTech SC has developed **eHealth**, an **Electronic Health Record System** aimed at modernizing the healthcare sector. This system digitizes patient records, streamlining clinical workflows and improving the quality of healthcare delivery.

So the **software products** are:

- **eHealth**: A sophisticated health management system for clinics and hospitals, featuring Electronic Health Records (EHR), telemedicine support, and pharmacy management.
- **eTech's ERP**: An integrated Enterprise Resource Planning system that automates critical core business processes, such as HR, inventory, sales, and project management.
- **eTech's CRM**: A specialized Customer Relationship Management solution, particularly tailored for insurance companies to streamline customer data and interaction management.

- **e79.8**: An advanced e-commerce platform supporting online commerce operations.
- **eShare**: A comprehensive system designed for efficient Shareholder Management.
- **eQuiteba**: An online portal facilitating savings and credit operations for SACCOs (Savings and Credit Cooperatives).
- **eTechRide**: A dedicated application platform for ride-hailing services.

The Infrastructure & Telecom Solutions

eTech provides reliable, world-class IT infrastructure and telecommunication services. This encompasses the supply and implementation of core enterprise components, including high-performance enterprise servers, robust network security, and advanced datacenter solutions. All services in this pillar are delivered and supported by certified technical experts.

Cybersecurity & Empowerment Solutions

The company offers a full suite of enterprise-grade cybersecurity services designed for protection and proactive risk management. Offerings include awareness training, governance, risk management, and compliance (GRC) services, alongside comprehensive technical assessments like penetration testing (covering vulnerability assessments, network, web, and mobile application testing).

Information Technology Outsourcing (ITO) & Digital Innovation Hub (DIH)

eTech serves as a trusted partner for organizations looking to outsource their end-to-end IT requirements. Furthermore, as a Digital Innovation Hub (DIH), the company actively fosters a culture of innovation by providing crucial support services such as ideation workshops, technical training, mentorship programs, acceleration services, and technology consulting.

The technologies underpinning these services are chosen for enterprise-grade demands, likely involving robust backend languages like **Java** or **.NET** for complex systems, modern frontend frameworks, relational databases like **Oracle** or **PostgreSQL** and infrastructure tools like **Docker** and **Kubernetes** to support their DevOps and product development offerings.

2.3. main customers or the end users

eTech S.C. serves a prestigious and diverse clientele across Ethiopia's key economic sectors. Its main customers include:

- **Healthcare Organizations:** Public and private general hospitals, referral hospitals, and specialized clinics, primarily utilizing the eHealth platform.
- **Savings and Credit Cooperatives (SACCOs):** Organizations served via its dedicated eQuitaba platform.
- **Government Agencies and NGOs**: A range of public sector institutions and non-governmental organizations.
- **Financial Institutions**: Major banks, insurance companies (e.g., Awach Insurance S.C.), and various microfinance organizations.

- **Various Ethiopian Enterprises**: Including major brands in transportation, commerce, and other services.

END USERS

The ultimate end users of eTech's products are the employees, management, and customers of these client organizations. This includes roles such as doctors, nurses, bank tellers, insurance agents, SACCO members, and the general public interacting with services like eTechRide and e79.8.

2.4. Organizational Structure

Its structure is inferred as a modern, divisional model organized around core service lines and specialized competencies to ensure deep expertise and efficient project execution. The company is led by an Executive Leadership team including a Chief Executive Officer and Directors, who are responsible for setting the overarching strategic vision and corporate direction.

Reporting to this leadership are various specialized divisions each headed by a department lead or manager. These core divisions logically include a Software Development unit, an Infrastructure and Telecom division managing data centers and networks, a Cyber Security team ensuring the integrity of all solutions and a dedicated Quality Assurance department upholding product reliability. Furthermore, the company has dedicated Project Management Offices and vertical specific teams. Supporting these technical divisions are essential corporate functions including Human Resources, Finance, Administration and Marketing & Sales which provide the operational backbone for the entire organization. This structure effectively balances deep specialization within divisions with the cross functional collaboration necessary to deliver integrated technology solutions.

2.5.Workflows

The workflow at eTech SC for project execution is a structured multi stage process designed to ensure the delivery of high quality, secure and scalable solutions aligning with its stated values of excellence and customer focus. The lifecycle typically begins with Requirement Analysis and Consulting often initiated by the Digital Transformation Services team who work closely with clients to understand their business challenges and define detailed functional and technical specifications. This is followed by the Solution Design and Architecture phase where senior architects and leads create a comprehensive technical blueprint by selecting the appropriate technology stack and defining system architecture with a strong emphasis on security and scalability. The project then moves into the Software Development and Implementation phase where agile development teams, potentially utilizing DevOps practices, build the solution in iterative sprints with version control and continuous integration pipelines managed by specialized staff.

A critical parallel phase is Quality Assurance and Testing where the dedicated QA team conducts rigorous testing including unit, integration, system and user acceptance testing (UAT) to validate the solution against all requirements before launch. Upon successful testing the Deployment and Go-Live phase is managed by infrastructure and deployment specialists who execute a carefully planned rollout to the client's production environment. Finally, the workflow transitions into the ongoing Maintenance and Support phase by providing clients with technical support, system monitoring and iterative improvements, ensuring a long-term partnership and the sustained health of the delivered solution.

Part Three: Internship experience

3.1. Internship Department and Rationale

We conducted our internship within the **Software Development Department** at eTech SC, where we were assigned to a team dedicated to building innovative solutions using the company's preferred technology stack. Our placement in this specific department was directly aligned with our personal interest and academic background as a Computer Science student.

We have a strong passion for developing practical software solutions that address real-world challenges in Ethiopia and beyond. This internship provided the perfect platform to apply our developer skills to a meaningful project, fulfilling our ambition to contribute to problem solving through website and application development.

The company's commitment to impactful projects, such as the national ERP system and eHealth platform resonated with our career goals making this department an ideal environment for our professional growth.

3.2. Executed Workpieces and Tasks

Our primary project was the end-to-end development of a Exhibition Management System. A comprehensive web application designed to streamline the organization and execution of conferences. The tasks We executed throughout this project encompassed the entire software development lifecycle. We were responsible for designing and implementing the backend RESTful API using the Vert.x toolkit, which included creating endpoints for user authentication, conference CRUD operations, and exhibitor management.

On the frontend, We developed a dynamic and responsive user interface with Angular CLI by building components for user registration, admin dashboards and payment integration. We designed and managed the PostgreSQL database schema, ensuring efficient data relationships for entities like users, conferences, exhibitors, and attendees. Furthermore, We integrated the Stripe payment gateway for handling booth rentals and registration fees, and implemented critical features such as JWT-based authentication, role-based access control (Admin, Exhibitor, Attendee), and file upload capabilities for logos and documents.

Specific Modules Developed:

- User Management & Authentication
- Conference Management (CRUD operations)
- Exhibitor Registration & Management
- Attendee Management
- Floor Management
- Speaker Management
- Sponsor & Partner Management
- Product Catalog Management
- Payment Processing System
- Administrative Dashboard

3.3. System Development Tools and Techniques

Throughout the project, We utilized a modern and robust technology stack that aligns with eTech SC's focus on scalable and efficient systems.

Backend Technologies:

- **Vert.x Framework (4.5.7)**: Reactive, non-blocking Java toolkit
- **PostgreSQL**: Relational database management
- **JDBC Client**: Database connectivity
- **JWT Authentication**: Secure API access
- **BCrypt**: Password hashing
- **Jackson**: JSON serialization/deserialization
- **Commons FileUpload**: File handling
- **Vert.x Mail Client**: Email integration

Frontend Technologies:

- **Angular 20.1.0**: Modern web framework
- **TypeScript**: Type-safe JavaScript
- **Angular Material**: UI component library
- **Reactive Forms**: Data validation
- **Angular Router**: Navigation management
- **HTTP Client**: API communication
- **Stripe.js**: Payment processing

Development Tools:

- VS Code: Primary code editor
- Postman: API testing and endpoint validation
- PostgreSQL: Database management and testing

Modern Development Practices:

- RESTful API Design: Standardized web service architecture
- CORS Implementation: Cross-origin resource sharing
- Role-Based Access Control: Multi-user permission system
- Reactive Programming: Asynchronous, non-blocking operations
- Component-Based Architecture: Modular frontend development

3.4. Major Challenges and Problems

We encountered several significant challenges, both technical and logistical.

Technical Challenges:

Framework Learning Curve:
- Angular Complexity: Understanding vast Angular framework ecosystem
- Vert.x Framework: Learning reactive programming concepts
- Integration Issues: Connecting frontend and backend systems

Development Obstacles:
- Bug Resolution: Code errors taking 20 minutes to over a week and Required intensive debugging
- CORS Configuration: Cross-origin request handling
- Naming Conviction errors

Hardware Issues:
- PC Breakdown: Our Computer fail during project development
- Financial Impact: Additional expenses for replacement equipment

Non-Technical Challenges:

Our non-technical challenges were predominantly logistical. Transportation was a major hurdle; Our 10+ km commute required us to wake up before 5:00 AM and often endure crowded buses, not arriving home until after 8:00 PM. This, coupled with financial constraints related to food and the unexpected burden of replacing Our personal laptop after it broke down, created a stressful environment. Furthermore, understanding the dynamics of a corporate business environment and how software projects align with real-world business expansion posed an initial conceptual challenge.

Transportation Difficulties:
- Distance
- Early Schedule: Waking up before 5:00 AM
- Transportation Timing: Arriving at station by 5:30 AM
- Return Journey: 2.5-3 hours daily travel time (5:30 PM - 8:30 PM)
- Standing Commute: No seating available on buses

Financial Constraints:
- Requesting money for transportation and food
- Equipment Costs: Additional PC purchase burden
- Daily Expenses: Managing food and transportation costs

Business Understanding:
- External Business Operations: Learning how companies function
- Real-world Application: Understanding practical business expansion
- Industry Dynamics: Grasping corporate work environments

3.5. Measures Taken to Overcome Challenges

Technical Problem Resolution Methodology

Our approach to software development challenges was structured and iterative, focusing on efficiency, stability, and continuous learning. These strategies were key to accelerating our skill acquisition and ensuring successful feature implementation.

Core Learning and Knowledge Acquisition Strategies

To bridge gaps between theoretical knowledge and the specific demands of the project, We employed the following strategies to proactively acquire necessary technical information:

- Peer Collaboration (Knowledge Sharing): We actively engaged in peer collaboration, seeking input from senior developers and fellow interns. These discussions provided exposure to alternative design patterns, allowed for quick validation of architectural choices and fostered a team-based approach to problem-solving.
- Documentation Research: We tried to read the official framework and library documentation (e.g., Angular, Vert.x). This methodical approach ensured best practices, established a primary source of truth for syntax and architecture, and facilitated a deeper understanding of the tool's intended use and capabilities.
- Online Resources : For specific error messages, obscure bugs, or understanding unique platform behaviors, We leveraged targeted Google searches and developer communities (like Stack Overflow). This allowed for rapid resolution of edge cases by accessing solutions tested by the wider development community, significantly reducing resolution time.
- Alternative Approaches: When a conventional method proved inefficient or overly complex, We practiced thinking of alternative ways to implement features. This involved evaluating architectural trade-offs, testing different library integrations, and ultimately choosing the path that offered the best balance of performance, maintainability, and code clarity.

Debugging Techniques:

- Systematic Approach: Methodical error identification
- Community Support: Leveraging developer networks
- Iterative Testing: Continuous validation using Postman
- Code Review: Collaborative problem-solving sessions

Non-Technical

Managing our daily life and energy was just as important as writing code. We used these simple steps to stay productive:

- Staying Focused: We kept our concentration high by making sure We had coffee and water. We took short walks outside to refresh our mind, talked with supportive friends when We needed a break to keep our energy up all day.
- Handling Travel: We planned our trips early, especially for morning work. We always had backup routes. We also used our travel time to plan our workday.
- Managing time

Practical Skills Development

Backend Development:
- Vert.x Framework: Reactive Java programming
- RESTful API Design: Standardized web service creation
- Database Management: PostgreSQL schema design and optimization
- Authentication Systems: JWT implementation and security
- CORS Configuration: Cross-origin request handling
- File Upload Systems: Image and document management

Frontend Development:
- Angular Framework: Modern web application development
- TypeScript Programming: Type-safe JavaScript development
- Component Architecture: Modular UI development
- Reactive Forms: Advanced form validation and handling
- Material Design: Professional UI implementation

Development Tools and Integration Skills

- **Postman Mastery:** Expertise in using Postman, an API platform for building and testing Application Programming Interfaces (APIs). Mastery involves creating, sending, and automating HTTP requests (GET, POST, PUT, DELETE) to backend endpoints. It includes validating responses, testing API reliability under different conditions, organizing requests into collections, and automating workflows to ensure the backend logic functions correctly before frontend integration.

- **Database Tools (PostgreSQL):** Competence in using PostgreSQL, a powerful, open-source object-relational database system.
- **Payment and email system integration**

3.6.Theoretical Knowledge Enhancement

Our internship at eTech SC served as a comprehensive deep dive into modern full-stack development, fundamentally enhancing my theoretical knowledge and practical skill set. We gained a robust understanding of core Web Development Principles, mastering the design of RESTful API Architecture with its standard HTTP Methods (GET, POST, PUT, DELETE) for CRUD operations, utilizing JSON for seamless data exchange, and implementing CORS Policies to manage secure cross-origin requests. This was fortified by hands-on experience with critical Security Concepts where We implemented JWT Authentication for token-based user sessions, enforced Role-Based Access Control (RBAC) to manage multi-level permissions across Admin, Exhibitor, and User roles, secured sensitive data using BCrypt for password hashing, and developed Protected Routes on both the frontend and backend.

Furthermore, We applied key Software Architecture patterns, including the MVC Pattern for a clean separation of concerns, embraced Reactive Programming with Vert.x for efficient asynchronous processing, utilized Component-Based Design in Angular for modular and reusable frontend development, and adopted an API-First Development approach to ensure a well-defined and scalable service-oriented architecture. This holistic experience has successfully bridged the gap between academic theory and industry standard practice.

3.7.Team Collaboration Skills

Collaborative Environment

Company Culture: The shared values, beliefs, and practices that shape the social and psychological environment of an organization. At eTech SC, this was characterized by a strong emphasis on community, mutual support, and collective growth, creating a workplace that felt more like a cohesive unit than just a collection of employees.

Family Atmosphere: A supportive and inclusive work environment where colleagues care for each other's personal and professional well-being. This was demonstrated at eTech SC through the genuine, family-like interactions among team members, where individuals supported one another beyond strict work duties, fostering a strong sense of belonging and psychological safety.

Tree Planting Activities: Organized community engagement initiatives where employees participate in activities that benefit the broader society. Participating in eTech SC's tree-planting events reinforced the values of teamwork, environmental responsibility, and contributing to a legacy beyond the office walls, strengthening our bond as a team.

Friday Coffee Ceremonies: Regular, traditional team-bonding sessions designed to break down formal barriers and foster informal interaction. These ceremonies at eTech SC were a cornerstone of the company culture, providing a dedicated space for employees and interns to connect on a personal level, share life experiences, and build relationships in a relaxed setting.

Equal Participation: An environment where every individual, regardless of their job title or seniority, is encouraged to contribute their ideas and opinions. During eTech SC's coffee ceremonies there was a clear absence of hierarchy, allowing interns to share freely alongside project managers and CEOs, ensuring all voices were heard and valued during our discussion at the coffee ceremony.

Communication Skills

- **Open Dialogue:** The practice of sharing both personal experiences and professional challenges openly within the team. At eTech SC, this was encouraged during coffee ceremonies and daily interactions, where conversations about life advice and career goals flowed freely, building trust and deep interpersonal connections.

- **Peer Learning:** A collaborative problem-solving approach where team members learn from and teach each other. When faced with bugs in our Vert.x or Angular code, We engaged in peer learning by discussing problems with other interns and developers, leading to shared solutions and a deeper collective understanding.

Mentorship: A guided learning relationship where experienced senior developers provide advice, knowledge, and support to less experienced colleagues. At eTech SC, We benefited greatly from the willingness of senior developers to explain complex concepts, review our code.

Team Dynamics

Supportive Network: A workplace environment where colleagues act as helpful friends, offering both technical and emotional support. The friends We made at eTech SC formed a supportive network that We could rely on when facing difficult challenges, both technical and personal, such as the stressful commute and complex coding problems.

Knowledge Sharing: A collaborative learning environment where information and skills are actively exchanged among team members. This was a key dynamic at eTech SC, where developers readily shared insights about frameworks, best practices, and solutions to common problems, accelerating everyone's growth.

Mutual Respect: A fundamental attitude where every team member is treated as an equal and valuable contributor.

Collective Problem-Solving: A team-based approach to debugging and overcoming obstacles, where multiple perspectives are leveraged to find a solution.

3.8. Leadership Skills Development

Project Ownership

- **Independent Decision Making:**The autonomy and responsibility to make critical technical and design choices for a project. During our internship, We exercised this by selecting appropriate libraries, designing specific database schemas for the conference system, and determining the implementation approach for features like the payment gateway integration, taking full ownership of the outcomes.
- **Problem-Solving Leadership:** The act of taking charge in resolving complex technical issues. When faced with persistent bugs We led debugging sessions by systematically breaking down the problem, researching solutions and coordinating with peers to test hypotheses until a resolution was found.
- **Technical Guidance:**The role of providing advice and support to colleagues on technical challenges. As We became more proficient with Vert.x and Angular, We actively helped other interns who were struggling with similar concepts, explaining debugging techniques and sharing resources that We had found useful.
- **Project Management:**The process of planning, executing, and overseeing a project from inception to completion. For the Exhibition Management System, We managed our own development timeline, set personal milestones, and ensured deliverables were met on schedule, effectively acting as the project manager for our own internship assignment.

Communication Leadership

- **Knowledge Sharing:** The practice of articulating and explaining complex technical concepts to peers in an accessible manner. We regularly engaged in this by breaking down how We implemented JWT authentication or resolved CORS issues, helping to elevate the entire team's understanding.
- **Mentoring Others:**The informal role of guiding and supporting less experienced colleagues. We naturally fell into a mentoring role with other interns, offering

help with their challenges, reviewing their code, and providing encouragement based on our own experiences overcoming obstacles.

- **Presentation Skills:** The ability to clearly demonstrate project progress, technical details, and outcomes to an audience. We honed this skill by presenting our work on the Conference Management System to our supervisors effectively communicating the functionality, architecture, and challenges of the project.
- **Cross-team Collaboration:** The skill of facilitating communication and cooperation between different departments or teams.

Initiative Taking

Proactive Problem-Solving: The practice of identifying potential issues or bottlenecks before they become critical and addressing them early. We demonstrated this by anticipating integration challenges between the frontend and backend and setting up comprehensive API tests in Postman to catch errors during development.

- **Quality Assurance:**The personal commitment to ensuring the quality, reliability, and performance of one's own work. We took ownership of this by rigorously testing our own code, writing unit tests for critical functions, and performing thorough checks before considering a feature complete, thereby ensuring the overall system's stability.

3.9.Work Ethics and Company Psychology

Our internship at eTech SC provided a profound immersion into professional work ethics and organizational psychology, shaping our understanding of a corporate environment.

Professional Development:

Work Ethics Understanding:

- **Punctuality**: Managing early morning schedules and long commutes
- **Dedication**: Working late nights to meet project deadlines
- **Quality Focus**: Ensuring thorough testing and debugging
- **Continuous Learning**: Seeking knowledge from documentation and peers

Company Culture Appreciation:

- **Inclusive Environment**: Equal participation regardless of hierarchy
- **Community Engagement**: Tree planting and social responsibility
- **Knowledge Sharing**: Friday coffee ceremonies for experience exchange
- **Professional Growth**: Learning from industry experts and CEOs

Business Understanding:

- **Real-world Application**: Understanding how companies operate
- **Industry Dynamics**: Grasping corporate work environments
- **Professional Networks**: Building relationships with industry professionals
- **Career Development**: Learning from successful professionals' experiences

3.10. Entrepreneurship Skills

Our internship at eTech SC was instrumental in cultivating a foundational entrepreneurship skill set, moving beyond pure coding to a holistic product-building mindset

Problem-Solving Mindset:

- **Solution-Oriented Thinking**: Developing systems to solve real problems
- **Innovation Focus**: Creating efficient conference management solutions
- **User-Centric Design**: Understanding end-user needs and requirements
- **Market Understanding**: Recognizing business opportunities in event management

Technical Entrepreneurship:

- **Full-Stack Development**: Comprehensive solution development
- **Scalable Architecture**: Designing systems for growth and expansion
- **Integration Capabilities**: Connecting multiple systems and services
- **Performance Optimization**: Ensuring efficient and reliable systems

Business Process Understanding:

- **Workflow Design**: Understanding business process automation
- **User Experience**: Creating intuitive and efficient interfaces
- **Financial Integration**: Implementing payment and transaction systems
- **Operational Efficiency**: Streamlining conference management processes

3.11.Interpersonal Communication Skills

Our internship at eTech SC was a transformative period for developing our interpersonal communication skills across technical, professional, and social domains. On a professional level, We cultivated strong team collaboration skills through daily interactions and development meetings, engaged in mentorship communication by both learning from senior developers and explaining concepts to fellow interns, and gained experience in cross-functional interaction, which deepened our client understanding by learning to interpret user requirements and feedback for the conference system. Socially, Our participation in the company's coffee ceremonies provided a unique platform to engage with the organizational culture, where We practiced peer support by offering both technical and emotional encouragement to colleagues, contributed to community engagement through tree-planting initiatives, and began building a professional network by forming relationships with industry experts, thereby strengthening our ability to connect and communicate in a multifaceted professional environment.

3.12. Recommendations and Conclusion

In conclusion, Our internship at eTech SC was a period of immense professional and personal growth, culminating in significant key achievements across technical, professional, and personal domains. Our primary technical accomplishment was the successful end-to-end development of a comprehensive Exhibition Management System, through which We mastered modern web development technologies including Angular, Vert.x and PostgreSQL. We successfully implemented critical, production-grade features such as secure JWT authentication and Stripe payment processing, all while adhering to principles of scalable and maintainable code architecture.

This hands-on experience was the catalyst for profound professional growth We developed robust problem-solving and debugging skills, significantly enhanced our team collaboration and communication abilities, gained a crucial understanding of real-world business operations, and began building a valuable professional network.

On a personal level, We navigated considerable logistical challenges to improve our time management and resilience, emerging with greater confidence in our abilities. Through full project ownership, We cultivated nascent leadership qualities and strengthened our interpersonal skills. This internship has not only solidified our passion for development but has also equipped us with a comprehensive skill set and a professional mindset, providing an invaluable foundation for our future career. We are deeply grateful to eTech SC for this transformative opportunity.

Part Four: Project Work

4.1.Summary of the Project

The Exhibition Management System developed during our internship at eTech SC is a full-stack web application that digitizes the end-to-end lifecycle of organizing and running conferences and exhibitions. It centralizes core operations conference scheduling, exhibitor registration and booth payments, attendee management, speaker coordination, sponsor/partner management, floor planning, and administrative oversight into a secure, scalable platform. The frontend is built with Angular (20.1.x) for a responsive user experience; the backend uses Vert.x (4.5.7) for high performance, reactive APIs; the database is PostgreSQL with a well-structured schema. payments are processed via Stripe and transactional and notification emails are supported via Vert.x Mail Client. The system enforces JWT-based authentication and role-based access control (Admin, Exhibitor, Attendee) and includes file upload support and optimized endpoints designed around REST principles. This project delivers measurable value by automating manual processes, improving data integrity, reducing operational overhead and enabling transparent, secure financial flows.

4.2.Background of the Project

The development of the Exhibition Management System was initiated to address significant inefficiencies inherent in traditional event planning. Previously, manual, spreadsheet driven processes were the norm, leading to frequent errors, substantial delays and poor visibility for organizers. Critical functions like exhibitor payments and booth allocation were managed through fragmented tools, which complicated tracking and increased the administrative effort required for reconciliation. Furthermore, attendee and speaker coordination was plagued by issues such as double-booking and a lack of clarity on session capacities. Communication including welcome messages, payment confirmations, event reminders, and password resets was entirely manual, lacking any form of automation. To resolve these challenges, the project was launched with the objective of building a single, scalable, secure and user-friendly platform, deliberately aligned with eTech SC's robust technology stack of Vert.x, Angular, and PostgreSQL.

4.3. Objective of the Project

4.3.1 General Objective

Design and implement a secure, scalable and maintainable Exhibition Management System that streamlines planning, registration, exhibitor workflows, payments, communications and administrative operations.

4.3.2 Specific Objectives

- Build CRUD modules for conferences, exhibitors, attendees, speakers, sponsors, partners, floors, and products.
- Implement JWT authentication, role-based access, and protected routes.
- Integrate Stripe for exhibitor booth payments with status tracking.
- Automate transactional emails (welcome, confirmations, password reset).
- Design a normalized PostgreSQL schema with indexes, triggers, and views.
- Provide an admin dashboard for user/role management and analytics stubs.
- Ensure responsive, accessible UI and secure file uploads.

4.4 Statement of the Problem for the Project

Existing processes are fragmented (spreadsheets, ad-hoc emails, manual payments), leading to data inconsistency, operational inefficiency, security risks and poor participant experience. There is no single source of truth for events, payments, capacity, or communications. The proposed system replaces the manual workflow with an integrated platform that enforces data integrity, security, auditability and scalability.

4.5 Methodology

- **Fact-finding:** This initial phase involved gathering critical information through direct consultation with the company advisor,conducting a comparative review of competing event platforms to identify industry standards and opportunities for innovation.

- **Iterative Development:** Instead of building the entire system at once, the project followed an iterative approach. This involved prioritizing high impact modules for initial development, followed by frequent demonstrations to stakeholders to gather feedback and make continuous improvements in subsequent cycles.
- **API-first:** This design strategy prioritized the database and backend infrastructure. Before any frontend development began, the API endpoints, data contracts (request/response formats) and authentication mechanisms were fully defined.
- **Continuous Testing:** To maintain high quality and reliability, testing was integrated throughout the development lifecycle.

4.6.Development Tools

IDE/Utilities: VS Code, Postman, Angular CLI, Maven.
Frontend: Angular 20.1.x, TypeScript, Angular Material, Reactive Forms, Router, HttpClient, Stripe.js.

Backend: Java 17, Vert.x 4.5.7 (web, auth-jwt, JDBC), Jackson (JSR-310), BCrypt, Commons FileUpload, Vert.x Mail Client.
Database: PostgreSQL (FKs, indexes, triggers, views).
Infrastructure (dev): Local environment with CORS enabled between Angular dev server and Vert.x API.

4.7 Functional and Non-Functional Requirements

4.7.1.Functional Requirements

Functional requirements define the specific behaviors and operations that a system must perform. For the Exhibition Management System, these were the core functionalities implemented:

4.7.1.1 User Authentication & Authorization

Secure user login and role-based access control, utilizing JWT for authentication and defining permissions for three distinct user roles: Admin, Exhibitor, and Attendee.

4.7.1.2 Conference Management

Full lifecycle management of conference events, including the ability to create, list, update, and delete conferences. This includes validating scheduling conflicts, managing attendee capacity, and tracking the active/inactive status of events.

4.7.1.3 Exhibitor Management

A streamlined exhibitor registration process for booth selection based on size and price, integrated Stripe payment processing, and functionality for exhibitors to upload their company logo.

4.7.1.4 Attendee Management

A system for attendees to register for conferences

4.7.1.5 Speaker Management

Creation and management of speaker profiles.

4.7.1.6 Sponsor & Partner Management

Maintenance of sponsor and partner profiles and the ability to associate them with specific events to acknowledge their support.

4.7.1.7 Floor Management

Definition of multi-floor venue layouts, setting capacity for each floor, and mapping exhibitors to their assigned booth locations.

4.7.1.8 Product Catalog Management

Full CRUD (Create, Read, Update, Delete) operations for exhibitors to manage their product catalogs, including the ability to upload product image

4.7.1.9 Email Automation

Automated sending of system-generated emails for user welcome, payment confirmation, password reset requests, and other essential notifications.

4.7.1.10 Administrative Dashboard

A comprehensive admin panel for user and role management, foundational stubs for system analytics and log viewing, and tools for content moderation.

4.7.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of a system, describing how the system should operate rather than what it does. For the Exhibition Management System, the following non-functional requirements were critical:

4.7.2.1 Security

The system was architected with a robust security posture, implementing JWT for secure session management, BCrypt for hashing passwords, comprehensive input validation to prevent injection attacks, CORS policies to control cross-origin requests, and security measures for guarded file uploads.

4.7.2.2 Performance

To ensure responsiveness under load, the backend was built using Vert.x for non-blocking I/O operations, database queries were optimized with strategic indexing, and stateless sessions were used to avoid server-side storage overhead.

4.7.2.3 Reliability

Data integrity and system dependability were achieved by leveraging PostgreSQL's ACID (Atomicity, Consistency, Isolation, Durability) transaction properties, alongside the use of database constraints and triggers to enforce business rules at the data layer.

4.7.2.4 Usability

The user interface was designed to be responsive across different devices, forms were built to be accessible, and the system provided consistent and clear feedback to users for all actions.

4.7.2.5 Scalability

The architecture was designed for future growth, allowing for the horizontal scaling of the stateless API, employing database indexing for query performance at scale, and having CDN-ready static assets for efficient content delivery.

4.7.2.6 Maintainability

The codebase was structured for long-term manageability through modular controllers and components in both frontend and backend, the use of environment-specific configuration files, and clear API contracts between services.

4.7.2.7 Observability (Baseline)

Foundational measures for system monitoring were put in place, including structured logging for easier debugging and analysis, with an architecture that is extensible for future integration of metrics collection and distributed tracing systems.

4.8 Use Case and Class Diagram

4.8.1 Use Case Diagram

Fig.4.8.1: Use case diagram of the system

Use Case Table

Use Case Id	Use Case Name	Actors	Description
UC-01	Login/Logout	Admin, Exhibitor	Allows authenticated users to securely access and exit the system.
UC-02	Register(Attendee/Exhib)	Guest	Allows a guest to create a new account as either an Attendee or an Exhibitor.
UC-03	Manage Conferences	Admin	Enables the admin to create, view, update, and delete conference events.
UC-04	Manage Users/Roles	Admin	Allows the admin to manage user accounts and assign system roles.
UC-05	Manage Exhibitors	Admin	Enables the admin to review, approve, and manage exhibitor registrations and booth assignments.
UC-06	Register Company /Booth	Exhibitor	Allows an exhibitor to complete their company profile and select a booth size, triggering the payment process.
UC-07	Process Payment	Exhibitor, System	Handles the secure processing of booth payments via the Stripe gateway.
UC-08	Manage Products	Exhibitor	Allows an exhibitor to create, read, update, and delete products in their catalog.
UC-09	Manage Content (Speakers, Sponsors, Partners)	Admin	Enables the admin to manage the profiles of speakers, sponsors, and partners associated with conferences
UC-10	Upload Media	Admin, Exhibitor	Allows users to upload logos (Exhibitor, Sponsor, Partner) and product images (Exhibitor).
UC-11	Browse Conferences & Sessions	Attendee, Guest	Allows users to view available conferences and their session details.

UC-12	Enroll in Sessions	Attendee	Allows an attendee to register for sessions within a conference.
UC-13	Receive Emails	Attendee, Exhibitor	The system automatically sends notifications (welcome, payment confirmation, password reset).
UC-14	View Analytics	Admin	Provides the admin with insights into system usage, registrations, and Recent Activities.
UC-15	Password Reset	Admin, Exhibitor	Allows a user to securely reset a forgotten password via email.

Table 4.8.1 usecase description

4.8.2. Class Diagram

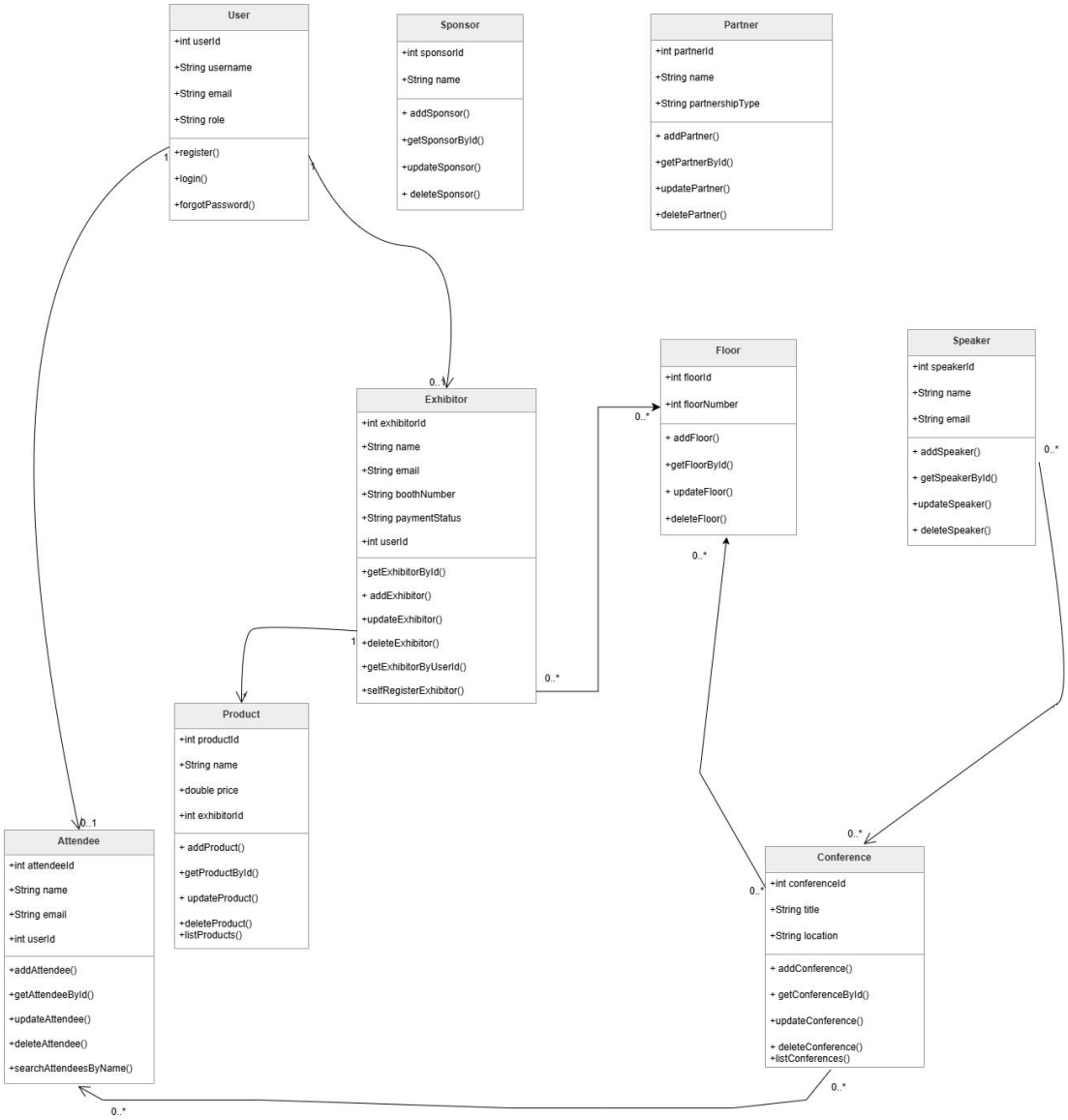

Fig 4.8.2 class diagram

Class description table

Class Name	Attributes	Methods	Description
Attendee	attendeeId(int), name(String), email(String), phone(String), registrationDate (date)	addAttendee(), getAttendeeById(), listAttendees(), updateAttendee(), deleteAttendee(), searchAttendeesByName()	CRUD and name search for attendee records.
User	userId(int), username (String), email(String), password (String), role(String),	register(),login(), forgotPassword(), listUsers(), updateUserRole(), deleteUser(),	Account identity, authentication, password flows, and

	resetToken (String), resetTokenExpiry (Long)	updatePassword(), changePassword()	admin user management.
Exhibitor	exhibitorId, name, contactPerson, email, boothNumber, logo,phone, category, description	listExhibitors(), getExhibitorById(), addExhibitor(), updateExhibitor(), deleteExhibitor(), updateExhibitorProfile(), selfRegisterExhibitor()	Exhibitor directory, self-registration, profile by user.
Product	productId (Integer),name, description, exhibitorId, category,price, image, lastModified	addProduct(), listProducts(), getProductById(), updateProduct(), deleteProduct()	Product catalog tied to an exhibitor.
Conference	conferenceId, title, description, startDate, endDate, location,capacity, status	addConference(), listConferences(), getConferenceById(), updateConference(), deleteConference()	Conference schedule and metadata.
Speaker	speakerId,name, bio, image	addSpeaker(), listSpeakers(), getSpeakerById(), updateSpeaker(), deleteSpeaker()	Speaker directory and maintenance.
Floor	floorId, floorNumber, floorName, capacity, description, isAccessible	addFloor(), getFloorById(), listFloors(), updateFloor(), deleteFloor()	Venue floor definitions.
Sponsor	sponsorId,name, contactPerson, contributionAmount, benefits	addSponsor(), getSponsorById(), listSponsors(), updateSponsor(), deleteSponsor()	Sponsor records and public listing.
Partner	partnerId,name, contactPerson, partnershipType, benefits, image	addPartner(), getPartnerById(), listPartners(), updatePartner(), deletePartner()	Partner records and public listing.

Part Five: Conclusion & Recommendation

5.1. Conclusion

Our internship period at **eTech S.C**, which lasted for two months and seventeen days, has been a highly meaningful and transformative journey for both of us. This experience allowed us to bridge the gap between the theoretical knowledge we gained at university and the practical skills required in a real world software development environment. Throughout the internship, we had the opportunity to actively participate in the development process, starting from understanding system requirements to contributing to the design and implementation of different system components. Working on real projects helped us gain valuable insight into how professional software systems are built, tested, and maintained.

During our time at eTech S.C., we were exposed to modern technologies and development tools, including Vert.x, Angular, and PostgreSQL, which significantly strengthened our technical abilities. We learned how to write cleaner code, debug issues efficiently, collaborate through version control systems, and follow structured development workflows. Beyond technical skills, this internship also helped us grow professionally by improving our time management, teamwork, communication, and problem-solving abilities. Facing unfamiliar technologies and complex tasks initially felt challenging, but over time, these challenges became opportunities for learning and growth, increasing our confidence in handling real development tasks.

Another important aspect of our internship experience was the collaborative environment we were part of. Working alongside experienced developers and supportive teammates allowed us to learn not only from formal instructions but also from daily discussions, feedback, and shared problem-solving sessions. We also gained a better understanding of workplace discipline, responsibility, and the importance of meeting deadlines while maintaining quality standards. These lessons have prepared us to adapt more easily to professional environments in the future.

We are deeply thankful to **eTech S.C.** for providing us with such a supportive and motivating learning environment. The mentorship and continuous guidance we received played a vital role in helping us expand our technical knowledge and develop professional confidence. This internship has strengthened our passion for software development and inspired us to continue improving our skills and pursuing excellence in our future careers.

Finally, we would like to express our sincere appreciation to our company supervisor, our intern friends, and the entire eTech S.C. organization for their patience, encouragement, and continuous support throughout this important stage of our academic and professional journey. The knowledge, experience, and confidence we gained during this internship will remain valuable assets as we move forward into our professional careers.

5.2 Recommendation

Based on the experience we gained during our internship, we would like to share several recommendations for different groups that play an important role in improving internship quality and student preparation. These suggestions are drawn from the practical challenges we faced and the lessons we learned throughout our training period.

5.2.1 Recommendations for Future Interns

From our experience, future interns should take initiative in their learning process rather than waiting to be directed at every step. Showing curiosity, asking thoughtful questions, and seeking clarification when something is unclear can significantly speed up understanding and skill development. When assigned tasks are completed, interns should look for ways to refine their work or volunteer to assist with new responsibilities, as this demonstrates commitment and enthusiasm.

It is also highly advisable for interns to maintain a daily or weekly record of their activities, newly learned tools, and solutions to problems they encounter. Keeping such notes makes it easier to prepare internship reports and also serves as a useful reference during job interviews or future projects.

Another important recommendation is to stay patient and persistent when facing unfamiliar tools or complex systems. Learning new technologies may initially feel difficult, but consistent effort and practice lead to noticeable improvement over time. In addition, interns should try to communicate with employees from different roles within the organization, such as testers, designers, or system administrators. This helps them understand how various departments work together and gives them a broader view of how software companies operate.

5.2.2 Recommendations for the Hosting Company (eTech S.C.)

We sincerely appreciate the opportunity given to us by eTech S.C., and based on our experience, we suggest a few improvements that could make future internship programs even more effective. Introducing a clearer orientation process at the beginning of the internship would help new interns quickly understand company policies, development tools, and project expectations. Providing a step-by-step introduction to commonly used systems and workflows could reduce confusion during the early stages.

In addition, scheduling regular short meetings between mentors and interns would be highly beneficial. These meetings would create opportunities for reviewing completed work, addressing difficulties, and receiving consistent feedback. Regular communication helps interns stay on track and feel more supported during their learning process.

Allowing interns to briefly observe or participate in different ongoing projects could also enhance their exposure to diverse technologies and development styles. This kind

of rotation would help interns gain wider practical knowledge and better understand the full range of activities within the company.

5.2.3 Recommendations for the Computer Science Department

From an academic perspective, we recommend that the Computer Science department continue strengthening practical learning opportunities that reflect current industry standards. Including more hands-on exercises using modern development frameworks and tools would help students become more comfortable when entering professional environments.

We also believe that increasing focus on teamwork, communication, and project organization skills would be highly valuable. Many real-world software projects require collaboration, clear communication, and proper planning, so students should be trained to work effectively in teams and manage tasks efficiently.

Furthermore, providing students with more guidance on technical documentation would be very helpful. Learning how to prepare structured documents such as system descriptions, design outlines, and user manuals would prepare students for real project environments where documentation is essential.

5.2.4 Recommendations for the University

At the university level, maintaining strong relationships with technology companies is essential for creating meaningful internship opportunities. Expanding partnerships with organizations similar to eTech S.C. would allow more students to gain valuable industry exposure and practical experience.

We also suggest organizing preparatory sessions before students begin their internships. These sessions could focus on workplace behavior, professional communication, and interview preparation. Such preparation would help students feel more confident and ready when entering professional environments.

Additionally, creating opportunities for experienced graduates to share their insights with current students could be highly beneficial. Hearing real life experiences from alumni working in the software industry would provide useful guidance, motivation, and practical advice for students preparing for internships.

6. References

The tools, technologies, and standards used throughout this internship were based on official documentation, widely accepted development practices, and industry-standard software tools.

- Angular Documentation: https://angular.io/docs
- Vert.x Documentation: https://vertx.io/docs
- PostgreSQL Documentation: https://www.postgresql.org/docs
- Postman: https://www.postman.com
- pgAdmin: https://www.pgadmin.org
- REST API principles based on HTTP standards
- IEEE software documentation standards (SRS guidelines)
- Freecodecamp.org : https://www.youtube.com/@freecodecamp